Ashmit Mukherjee

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New York University Abu Dhabi

Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science; minor in Economics. Cumulative GPA: 3.599/4.000
Study away: NYU Paris and NYU New York
Selected coursework: Social Networks; Natural Language Processing; Principles of Data Science I; Statistics for the Social and Behavioral Sciences; Algorithms

Research Interests

Computational social science, experimental methods, and machine learning. Interested in how computational tools can help investigate social, economic, and historical questions; current work concerns human-AI interaction and experimental systems.

Research Experience

Research Assistant, Human-AI Collaboration *Feb 2026 - present*
NYU Abu Dhabi | With Ben Rosche and Hanan Salam

- Lead the technical implementation of an online platform for a faculty-led human-AI research project; contribute to study design and translate research requirements into software behavior and data collection.
- Direct implementation and review across Python/FastAPI mechanics, Empirica participant workflows, session orchestration, and structured experimental exports.
- Develop requirements and checks for action validation, information boundaries, resumable computational runs, and independent reconstruction of outcomes.

Research Assistant, eBRAIN Lab *Feb 2026 - present*
NYU Abu Dhabi | With Muhammad Shafique and Minghao Shao

- Extended protein-model evaluation to signal-peptide classification and cleavage-site prediction, including ESM-2 task adapters, model heads, and metrics.
- Ran LoRA, full fine-tuning, and biological-prior comparisons on the university HPC cluster across 150M, 650M, and 3B models; investigated initialization, feature injection, and efficiency as distinct experimental choices.
- In a three-seed 650M follow-up, measured 86.08% mean exact cleavage-site match for BLOSUM62 feature injection versus 83.64% for vanilla LoRA on 1,231 signal-peptide-positive examples. Full fine-tuning led this metric at 150M and 3B.
- Used repeated runs and prior controls to test whether promising settings held up; two single-seed settings reversed on three-seed confirmation.

Research Manuscript

D. S. Abubaker, M. Faheem, **A. S. Mukherjee**, S. Othman, S. Sifar, S. Sthapit, Y. Wang, M. Shao, and M. Shafique.
Bio-LoRA: Biological Priors as Inductive Bias for Low-Rank Adaptation.
Unpublished manuscript, 2026. Co-first author.
Contribution and follow-up experiments

Independent Research

Synthetic Pilot Lab, Computer Science Capstone *Oct 2026 - present*
In development; first diagnostic specified

- Investigate whether synthetic participants can help find faults in experimental software before human piloting.
- Specified paired correct and faulty implementations, scripted trajectories that expose or avoid a known fault, and an independent evaluator that reconstructs expected states and payoffs.

Selected Projects

FieldLens, Evidence-linked Incident Review *Oct 2026*

- Directed a local-model prototype using fictional incident records; specified bounded retrieval, exact source references, and a deterministic fallback when model inference fails.
- Compared free-prose and constrained-claim answers on saved synthetic cases, separating output validity from relevance and completeness. Used failed and incomplete answers to direct revisions.

Hinglish Named-Entity Recognition *Fall 2024*

- Contributed training pipelines, evaluation tooling, and error analysis in a four-person project using mBERT and XLM-RoBERTa on COMI-LINGUA.
- Evaluated encoders on the full 4,829-example test split and zero-shot language-model baselines on a separate 100-example sample; kept the comparison scopes explicit.

CAMP, Campus Asset Management Platform *Fall 2024*

- Built staff-workflow REST endpoints, alerts, and dashboard statistics in a five-person React, Express, and MongoDB project; configured backend integration testing and client API utilities.
- Served as Product Owner and Scrum Master in rotating team sprints.

Applied Experience

Machine Learning Engineer Intern, Zeek *Jun - Aug 2025*

- Built and evaluated customer-segmentation workflows; compared preprocessing and model choices and used SHAP to examine model behavior.

Market Research Consultant, Global Media Alliance Broadcasting *Jun 2024*

- Conducted audience analysis, survey design, and customer-engagement research; presented quantitative and qualitative findings to senior leadership.

Service

First Year Program Facilitator, NYU Abu Dhabi *Aug - Dec 2024*
Facilitated workshops, group activities, and weekly check-ins for more than 20 first-year students.
Front of House Staff, NYU Abu Dhabi Arts Center *Jan - May 2024*
Supported guest check-in, ticketing, and event operations for audiences exceeding 500 attendees.

Methods and Tools

Research: Experimental design, statistical modeling, model evaluation, reproducible experiment workflows
Programming and data: Python, JavaScript, C++, C#, pandas, NumPy, SciPy
Machine learning: PyTorch, Transformers, scikit-learn, LoRA, ESM-2, XGBoost, SHAP
Systems: FastAPI, Empirica, Slurm, Docker, pytest, Git, GitHub Actions, MLflow